Asmedia ASM1061
PCI-E to SATA Controller

Software Programming Note

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Revision 4.2

2019/09/16

ASMEDIA TECHNOLOGY INC.

Revision History

Rev	Date	Author	Description
4.2	2019/09/16	Jesse Chang	Add section 15. Disable 64-bit Addressing Capability
4.1	2018/05/08	Jesse Chang	Modify the description of payload size.
4.0	2016/12/19	Jesse Chang	1. Modify title to ASM1061 2. Re-typeset from revision 3.2 3. Add Feature Registers setting
3.2	2015/06/23	Jesse Chang	Set RxCAE=92h, RxDAE=92h
3.1	3014/02/06	Jesse Chang	Add “PCIE Max Payload Size Compatibility”
3.0	2013/06/21	Jesse Chang	Add “Enable L1 ASPM Supported”
2.1	2013/02/27	Jesse Chang	Set PCI RxF8 =1 for OPROM v2.x or later version backward-compatible v1.00 and 0.9x
1.9	2012/06/22		Enable PCIE L0s support for WHCK test
1.8	2012/02/16	Jesse Chang	Add SSC Control
1.7	2011/06/23	Jesse Chang	Add PCI Revision ID Compatibility
1.6A	2011/05/19	Jesse Chang	Disable PCIE L0s support
1.5	2011/03/31	Jesse Chang	Add Set Shadow RAM Cacheable
1.4	2011/03/02	Jesse Chang	1. Modify “PCIE Compatibility” to “PCIE Power Management Compatibility” and its contents. 2. Add RxCA4 and RxDA4 setting in “SATA Optimal Setting for Devices Compatibility”
1.3	2011/02/08	Jesse Chang	Add Internal Regulator Setting
1.2	2011/01/29	Jesse Chang	Set RxA01[0] = 0 to disable “PCIE L0s active state power management support” for PCIE compatibility.
1.1	2011/01/03	Jesse Chang	1. Modify subject title for item 5 Spin-Up SATA Device 2. Modify “ASM106x PCI-E SATA 3 Host Controller” to “ ASM106x Series PCI-E to SATA Host Controller”
1.0	2010/12/13	Jesse Chang	Initial release

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1. Introduction

Asmedia ASM1061 is a PCIE to SATA host controller and supports 2 SATA ports. This document describes how to program and which registers are needed to program by System BIOS when SPI flash ROM is not used. All registers setting can be done by SPI flash ROM if a SPI flash ROM is used with ASM1061.

2. Set PCI Subclass Code for SATA Controller

The default value of ASM106x SATA controller Subclass code is 01h (IDE) and can be modified by follow sequence:

1. Enable Subclass to be programmed

   Set PCI RXEC[2] = 1
2. Set Subclass Code

   Modify PCI RX0A according the following table
3. Disable Subclass to be programmed

   Set PCI RXEC[2] = 0

This table describes the relationship between PXI RX09, RX0A and RX0B.

Program Interface RX09	Subclass code RX0A	Base Class Code RX0B	Meaning
85h	01h	01h	IDE, Native PCI mode
01h	06h	01h	AHCI, Version 1.x
00h	04h	01h	RAID

Note: Don't Modify RX09 and Rx0B. Rx09 will be changed automatically while RX0A is modified. RX0B must remain 01h.

PCI Configuration Space offset – ECh

Register Name: MISC

Size: 8 bits

Bit	Attrib	Description	Default	Recommend
7:3	Rsvd	**Reserved** Don't program this field. Keep its value	0	
2	RW	**PCI Read-Only Registers Write Enable** 1: Write Enabled 0: Write Disabled PCI Subsystem Device ID, Subsystem Vendor ID, Subclass Code, and Base Class Code registers can be modified when set to '1'. This bit must be set to '0' after modified to keep read-only attribute.	0	

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1	RW	**Spin-up SATA Port 1** Set to '1' to spin-up SATA port 1 for IDE mode.	0	
0	RW	**Spin-up SATA Port 0** Set to '1' to spin-up SATA port 0 for IDE mode.	0	

PCI Configuration Space offset – 0Ah

Register Name: SUBCLASS

Size: 8 bits

Bit	Attrib	Description	Default	Recommend
7:0	RO	**Subclass Code**	01h	

3. Set Subsystem Vendor ID and Subsystem ID

The default values of SVID and SSID are 1B21h and 1060 and can be modified by follow sequence:

1. Enable SVID and SSID to be programmed

   Set PCI RXEC[2] = 1
2. Set SVID and SSID

   Modify PCI RX2C ~ RX2F to customer values
3. Disable SVID and SSID to be programmed

   Set PCI RXEC[2] = 0

Subclass Code, SVID and SSID can be modified simultaneously.

PCI Configuration Space offset – ECh

Register Name: MISC

Size: 8 bits

Bit	Attrib	Description	Default	Recommend
7:3	Rsvd	**Reserved** Don't program this field. Keep its value	0	
2	RW	**PCI Read-Only Registers Write Enable** 1: Write Enabled 0: Write Disabled PCI Subsystem Device ID, Subsystem Vendor ID, Subclass Code, and Base Class Code registers can be modified when set to '1'. This bit must be set to '0' after modified to keep read-only attribute.	0	
1	RW	**Spin-up SATA Port 1** Set to '1' to spin-up SATA port 1 for IDE mode.	0	
0	RW	**Spin-up SATA Port 0** Set to '1' to spin-up SATA port 0 for IDE mode.	0	

PCI Configuration Space offset – 2Ch

Register Name: SVID

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Size: 16 bits

Bit	Attrib	Description	Default	Recommend
15:0	RO	**Subsystem Vendor ID**	1B21h	

PCI Configuration Space offset – 2Eh

Register Name: SSID

Size: 16 bits

Bit	Attrib	Description	Default	Recommend
15:0	RO	**Subsystem Device ID**	1060h	

4. SATA Optimal Setting for Devices Compatibility

Following registers must be set for ASM1061 before spin up for better SATA devices compatibility.

1. Set PCI RxCA4 = ABh before Spin-Up for SATA Port 0
2. Set PCI RxCA5 = 26h before Spin-Up for SATA Port 0
3. Set PCI RxCAE = 92h before Spin-Up for SATA Port 0
4. Set PCI RxDA4 = ABh before Spin-Up for SATA Port 1
5. Set PCI RxDA5 = 26h before Spin-Up for SATA Port 1
6. Set PCI RxDAE = 92h before Spin-Up for SATA Port 1

PCI Configuration Space offset – CA4h, DA4h

Register Name: PHY24

Size: 8 bits

Bit	Attrib	Description	Default	Recommend
7:4	RW	**SATA Gen 1 Transmitter driver pre/de-emphasis or slew rate level control**	Ah	Ah
3:0	RW	**SATA Gen 1 Transmitter driver current control**	8h	Bh

PCI Configuration Space offset – CA5h, DA5h

Register Name: PHY25

Size: 8 bits

Bit	Attrib	Description	Default	Recommend
7:4	RW	**SATA Gen 2 Transmitter driver pre/de-emphasis or slew rate level control**	6h	2h
3:0	RW	**SATA Gen 2 Transmitter driver current control**	3h	6h

PCI Configuration Space offset – CAEh, DAEh

Register Name: PHYAE

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Size: 8 bits

Bit	Attrib	Description	Default	Recommend
7	RW	**CP current value fine tune select** pmode_bw asserted	1	1
6:4	RW	**Receiver PLL bandwidth select** pmode_bw asserted	3'000b	3'001b
3	RW	**CP current value fine tune select** pmode asserted	0	0
2:0	RW	**Receiver PLL bandwidth select** pmode asserted	3'010b	3'010b

5. Spin-Up SATA device

SATA device must be spun up before detected on ASM106x SATA controller. PCI RXEC[1:0] are spin-up device control. PCI RXEC[0] for SATA port 0 and PCI RXEC[1] for SATA port 1. These 2 bits must set 2'11b to spin up SATA devices.

For IDE mode, PCI RXEC[1:0] must be set. For AHCI mode, RXEC[1:0] is recommended to set but it is optional. AHCI mode has another spin-up bit in Port Command Register (PxCMD).

PCI Configuration Space offset – ECh

Register Name: MISC

Size: 8 bits

Bit	Attrib	Description	Default	Recommend
7:3	Rsvd	**Reserved** Don't program this field. Keep its value	0	
2	RW	**PCI Read-Only Registers Write Enable** 1: Write Enabled 0: Write Disabled PCI Subsystem Device ID, Subsystem Vendor ID, Subclass Code, and Base Class Code registers can be modified when set to '1'. This bit must be set to '0' after modified to keep read-only attribute.	0	
1	RW	**Spin-up SATA Port 1** Set to '1' to spin-up SATA port 1 for IDE mode.	0	
0	RW	**Spin-up SATA Port 0** Set to '1' to spin-up SATA port 0 for IDE mode.	0	

6. Internal Regulator Setting

ASM1061 can use external or internal regulator for PWM. When use internal regulator, the voltage of it must be set to 1.28v before Spin-Up devices. Set PCI RxA12[4:3] = 2'11b to adjust the voltage to 1.28v

Setting PCI RxA12[4:3] has no side effect and is unavailable If ASM1061 uses external regulator.

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So PCI RxA12[4:3] can be always set to 2'11b.

PCI Configuration Space offset – A12h

Register Name: PWMCTL

Size: 8 bits

Bit	Attrib	Description	Default	Recommend
7:5	Rsvd	**Reserved** Don't progrm this field. Keep its value	3'000b	3'000b
4:3	RW	**reference voltage** 00: -3.25% 01: 0% 10: 3.25% 11: 6.5%	2'01b	2'11b
2:0	Rsvd	**Reserved** Don't progrm this field. Keep its value	3'011b	3'011b

7. Feature Registers Setting

There are three feature registers: PCI RxFD, RxFE, and RxFF. They are described in following. These feature registers are strongly recommended to set for best system compatibility.

PCI Configuration Space offset – FDh

Register Name: Feature2

Size: 8 bits

Bit	Attrib	Description	Default	Recommend
7:1	Rsvd	**Reserved**	0	0
0	RW	**S4 wake up improved** 1: improve wake up time on Win 8	0	1

Only bit 0 can be programed and is recommended set to 1 to improve S4 wake-up time on Microsoft Window 8 operating system.

PCI Configuration Space offset – FEh

Register Name: Feature_1

Size: 8 bits

Bit	Attrib	Description	Default	Recommend
7:6	Rsvd	**Reserved** Don't program. Keep to 2'00b	2'00b	2'00b
5:4	RW	**ATAPI command improved** 11: Improve ATA and ATA command are executed simultaneously 00: ATAPI command is not improved Others: reserved	2'00b	2'11b
3:2	RW	**SATA 6G disabled**	2'00b	2'00b

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		11: Disable port 0 and port 1 SATA 6G 10: Disable port 1 SATA 6G only 01: Disable port 0 SATA 6G only 00: SATA 6G is not disabled		
1:0	RW	**PCIe ASPM L1 improved** 11: PCIe ASPM L1 improved 00: PCIe ASPM L1 improved Others: reserved	2'00b	2'11b

Bit 7:6 - Reserved bits. Don't set these two bits.

Bit 5:4 - Recommend to set 2'11b to ATA and ATAPI device are connected on ASM1061

Bit 3:2 - Disable SATA 6G support. It is not recommended to disable SATA 6G

Bit 1:0 - Must be set 2'11b if PCIe L1 active state supported is enabled (PCI RXA01[1] = 1)

PCI Configuration Space offset – FFh

Register Name: Feature_0

Size: 8 bits

Bit	Attrib	Description	Default	Recommend
7	RW	**HDD compatibility improved** 1: improve HDD compatibility	0	1
6:2	Rsvd	**Reserved** Don't program. Keep to 0h	00h	00h
1	RW	**AHCI port 1 PxCMD.ESP** 1: Set AHCI port 1 PxCMD.ESP bit	0	
0	RW	**AHCI port 0 PxCMD.ESP** 1: Set AHCI port 0 PxCMD.ESP bit	0	

Bit 7 - Recommend to set for HDD compatibility.

Bit 6:2 - Reserved bits. Don't set these bits..

Bit 1 - Modify AHCI port 1 PxCMD.ESP and HPCP bit.

Bit 0 - Modify AHCI port 0 PxCMD.ESP and HPCP bit.

When to program bit 0, 1, Please refer to Disable eSATA and Hot-plug for AHCI Mode.

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8. PCIe L0s/L1 Power Management Compatibility

ASM1061 L0s and L1 active state supported can be enabled/disabled by PCI RXA01[1:0].

1. **PCIe L0s active state:**

   L0s active state supported must be disabled in ASM1061 for PCIe compatibility. Set PCI RxA01[0] = 0 to disable. Microsoft WHCK will check the L0s supported bit in PCIE Link Capabilities Register (offset 0x0C bit 10), PCI RxA01[0] must be set 1 to pass WHCK test.

2. **PCIe L1 active state:**

   L1 active state supported must be disabled in ASM106x for PCIE compatibility. Set PCI RxA01[1] = 0 to disable. If L1 active state supported is enabled, PCI RXFE[1:0] must be set to 2'11b.

PCI Configuration Space offset – A01h

Register Name: ASPM

Size: 8 bits

Bit	Attrib	Description	Default	Recommend
7:4	Rsvd	**Reserved** Don't progrm this field. Keep its value	4h	4h
3:2	Rsvd	**Reserved** Don't program. Keep to 2'00b	2'00b	2'00b
1	RW	**PCIe ASPM L1 Supported** 1: L1 is supported 0: L1 is not supported	0	
0	RW	**PCIe ASPM L0s Supported** 1: L0s is supported 0: L0s is not supported	0	0

9. Disable eSATA and Hot-plug for AHCI Mode

The Hard Disk Drive (HDD) which connected on ASM1061 SATA port will be displayed on Windows Notification Area and can be safely removed when use Windows inbox driver. Windows inbox AHCI driver depends on AHCI HBA Capabilities register and each AHCI Port Command and Status register to determine whether displays the HDD on Notification Area.

When HBA Capabilities register bit 5 (CAP.SXS) is 1 that indicates the HBA supports External SATA. When Port Command and Status register bit 18 (PxCMD.HPCP) is 1 that indicates the SATA port supports Hot-Plug (please refer to AHCI specification). If CAP.SXS or PxCMD.HCPC is 1, Windows will display the HDD which connect on this AHCI SATA port. To eliminate HDD displayed on Notification Area, the HBA CAP.SXS and PxCMD.HCPC bit of every port must be set to 0.

AHCI specification defines that CAP.SXS and PxCMD.HPCP are read only bits. They can't be modified by software. ASM1061 provides other PCI registers to modify these AHCI read only bits - RxE5C[5] for modifying CAP.SXS, RxFF[0] for modifying Port 0 PxCMD.HCPC and RxFF[1] for modifying Port 1 PxCMD.HCPC. To disable eSATA and hot-plug for AHCI Mode, please set following registers.

PCI RxE5C[5] = 0 – Set HBA CAP.SXS = 0

RxFF[0] = 1 – Set Port 0 PxCMD.HPCP = 0

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RxFF[1] = 1 – Set Port 1 PxCMD.HPCP = 0

PCI Configuration Space offset – E5Ch

Register Name: AHCI_CAP

Size: 8 bits

Bit	Attrib	Description	Default	Recommend
7	RW	**Command Completion Coalescing Supported (CCCS)**	1	1
6	RW	**Enclosure Management Supported (EMS)**	0	0
5	RW	**Supports External SATA (SXS)**	1	
4:0	RW	**Number of Ports (NP)**	01h	01h

10. SSC Control

SSC can be enabled to reduce Electromagnetic Interference (EMI). PCI RxA10 can be set to control SSC. SSC mode is recommended to set to 0~-1250ppm. The SSC enable bit must be set at least 100 ns after other bits are set. Following is steps of enabling SSC.

1. Set PCI RxA10[4:2] = 2'011b, SSC mode 0~-1250ppm, 1:1 modulation
2. Delay 100 ns at least
3. Set RxA10[0] = 1, SSC enable

PCI Configuration Space offset – A10h

Register Name: SSC

Size: 8 bits

Bit	Attrib	Description	Default	Recommend
7:5	RW	**SSCPLL bandwidth** 000: 10uA 001: 20uA 010~111: reserved	0	
4:2	RW	**SSC mode** 000: 0~-4000ppm, 1:1 modulation 001: 0~-4000ppm, 1:3 modulation 010: 0~-5000ppm, 1:1 modulation 011: 0~-1250ppm, 1:1 modulation 1xx: -500~-4500ppm, 1:1 modulation	0	
1	RW	**SSC manual profile** 0: auto mode 1: manual mode	0	
0	RW	**SSC enable**	0	

11. PCIe Max Payload Size Compatibility

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PCIE Max Payload Size is setting on PCIe Device Control Register bit [7:5] and defined

3'000b 128 bytes max payload size

3'001b 256 bytes max payload size

3'010b 512 bytes max payload size

3'011b 1024 bytes max payload size

3'100b 2048 bytes max payload size

3'101b 4096 bytes max payload size

3'110b Reserved

3'111b Reserved

PCI Rx88 of ASM1061 is the PCIe Device Control Register. The default value of ASM1061 is 128 bytes max payload size (3'000b). Don't set this value greater than 256 bytes max payload size otherwise the data transfer between ASM1061 and PCIE root port will be incorrect.

PCI Configuration Space offset – 88h

Register Name: DCTL

Size: 8 bits

Bit	Attrib	Description	Default	Recommend
7:5	RW	**Max Payload Size**	3'000b	3'000b or 3'001b
4	RW	**Enable Relaxed Ordering**	0	
3	RW	**Unsupported Request Reporting Enable**	0	
2	RW	**Fatal Error Reporting Enable**	0	
1	RW	**Non-Fatal Error Reporting Enable**	0	
0	RW	**Correctable Error Reporting Enable**	0	

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13. System BIOS Program Time

All above registers must be set in System BIOS when cold boot, warm boot, S3 resume, S4 resume and before assigning PCI resources.

14. Set Shadow RAM Cacheable

Legacy OPROM occupied partial shadow RAM space when system POST but it depends on System BIOS. Please set shadow RAM which ASM1061 legacy OPROM occupied cacheable for improving performance in DOS. The HDD access performance will be great impact if shadow RAM is not cacheable. Example: the transfer speed from 3GB/m down to 300MB/m in ghost application.

15. Disable 64-bit Addressing Capability

AHCI driver uses 32-bits or 64-bits addressing for ASM1061 depending on HBA Capabilities register bit 31 (CAP.S64A). The 64-bit addressing capability can be disabled if there is problem. When 64-bit addressing is disabled, the upper 32-bits of PxCLB and PxFB must clear to zeros. Please set following registers to disable 64-bits addressing capability:

PCI RxE5F[7] = 0 – Set HBA CAP.S64A = 0
AHCI MMIO Rx104[31:0] = 00h – Set Port 0 CLBU = 00h
AHCI MMIO Rx10C[31:0] = 00h – Set Port 0 FBU = 00h
AHCI MMIO Rx184[31:0] = 00h – Set Port 1 CLBU = 00h
AHCI MMIO Rx18C[31:0] = 00h – Set Port 1 FBU = 00h

PCI Configuration Space offset – E5Fh

Register Name: AHCI_CAP3

Size: 8 bits

Bit	Attrib	Description	Default	Recommend
7	RW	**Supports 64-bit Addressing (S64A)**	1	
6	RW	**Supports Native Command Queuing (SNCQ)**	1	1
5	RW	**Supports SNotification Register (SSNTF)**	1	1
4	RW	**Supports Mechanical Presence Switch (SMPS)**	0	0
3	RW	**Supports Staggered Spin-up (SSS)**	1	1
2	RW	**Supports Aggressive Link Power Management (SALP)**	0	0
1	RW	**Supports Activity LED (SAL)**	1	1
0	RW	**Supports Command List Override (SCLO)**	1	1

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AHCI MMIO offset – 104h

Register Name: P0CLBU

Size: 32 bits

Bit	Attrib	Description	Default	Recommend
31:0	RW	**Port 0 Command List Base Address Upper 32-bits**		

AHCI MMIO offset – 10Ch

Register Name: P0FBU

Size: 32 bits

Bit	Attrib	Description	Default	Recommend
31:0	RW	**Port 0 FIS Base Address Upper 32-bits**		

AHCI MMIO offset – 184h

Register Name: P1CLBU

Size: 32 bits

Bit	Attrib	Description	Default	Recommend
31:0	RW	**Port 1 Command List Base Address Upper 32-bits**		

AHCI MMIO offset – 18Ch

Register Name: P1FBU

Size: 32 bits

Bit	Attrib	Description	Default	Recommend
31:0	RW	**Port 1 FIS Base Address Upper 32-bits**		

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